# Project Draft Guidelines

MAT 219, Fall 2026

Your draft should show what your group has done with the data. Build on your proposal. Reuse and revise your research question, motivation, data source, and variable descriptions, incorporating my feedback. Keep this information in the draft so it can be read on its own; you do not need to write it again from scratch. The main new work is your analysis, interpretation, limitations, and next steps.

## What to include

**Project title and group members**

Give your project a clear, descriptive title and list all group members beneath it.

**1. Research question and motivation**

Carry forward your research question and brief motivation from the proposal, revising them to address my feedback. If your question or dataset has changed, briefly explain the change.

**2. Data description**

Reuse your data source, working link, and descriptions of the variables relevant to your analysis; update them as needed. Focus on new details: what one row represents, the actual sample size used, missing values, and any exclusions, filtering, merging, or other data preparation.

**3. Exploratory analysis**

Use R to provide relevant summary statistics and at least two clearly labeled graphs that help answer your question(s). Explain what each graph or summary tells you; do not simply paste output. Summarize the main patterns you notice.

**4. Statistical methods and preliminary results**

Identify the confidence interval, hypothesis test, or regression method appropriate for each question and explain why it fits. Define the population quantity of interest; for a test, state the null and alternative hypotheses. Check relevant conditions and discuss concerns about independence or how the data were collected. Report available results and interpret them in context.

**5. Preliminary conclusions and limitations**

Explain what you can conclude so far and what remains uncertain. Discuss limitations of the data or methods, including whether your results can be generalized and whether a causal conclusion is justified.

**6. Next steps, group roles, and questions for feedback**

Include a short role allocation table with three columns: remaining task, responsible group member(s), and target completion date. Cover the work needed for the presentation and final report, such as data preparation, analysis and condition checks, interpretation and report revision, and slides and rehearsal. Include specific questions you would like feedback on.

The assigned member leads each task, but all group members must review and understand the methods, code, and results. Do not divide the project so that only one person understands the analysis. Roles may be adjusted as the project develops.

## What should be complete by the draft

Your data should be loaded and prepared in R, and your exploratory analysis should be complete. Attempt the statistical analysis using methods we have covered by November 16. If your project needs a method we have not yet covered, explain the planned method and what you will use it to investigate. Clearly label unfinished work and explain what you still need to do. Contact me before the deadline if you are unsure which method fits your question.

## Format and submission

- Aim for about 800-1,200 words, excluding graphs, tables, code, and references. Use the sections above to organize your draft.
- Submit one PDF draft per group on Canvas by November 16. List every group member on the document.
- Include relevant graphs and tables in the draft, with labels and explanations. Cite the dataset and any other sources used.
- Submit the R code used so far as a separate .rmd file. Include data loading, preparation, graphs, and analysis, with brief comments. Provide the data file(s), or a direct download link and clear loading instructions, so I can run your code.

## How the draft is graded

The draft is graded on the substance of your progress, your statistical reasoning, and the clarity of your explanations. Preliminary results may change after feedback; identify uncertainties honestly.

| **This will be graded on whether…** | ✓ |
|---|---|
| The research question is specific and proposal feedback has been addressed | ☐ |
| The data, variables, and preparation steps are clearly explained | ☐ |
| Relevant summaries and at least two graphs are included and interpreted | ☐ |
| Methods are justified, conditions are considered, and available results are interpreted | ☐ |
| Limitations, remaining tasks, a role allocation table, and questions for feedback are included | ☐ |
| The draft and commented R code are organized and submitted on time | ☐ |

## A note on using AI tools

Your draft should reflect your group's own understanding. If you use AI tools to help with code or explanations, check the results against your data and course materials. Every group member should be able to explain the methods, code, and conclusions submitted.